



Несовмест. $P(A \cup B) = P(A) + P(B)$
"или"

Совмест. $P(A \cup B) = P(A) + P(B) - \underline{P(A \cap B)}$

$P: > 2$ и четн.
4, 6 ←

$$P(A) = 3, 4, 5, 6 = \frac{4}{6} = \frac{2}{3}$$

$$P(B) = 2, 4, 6 = \frac{3}{6} = \frac{1}{2}$$

$$P(A \cap B) = 4, 6 = \frac{2}{6} = \frac{1}{3}$$

$$P(A \cup B) = \frac{2}{3} + \frac{1}{2} - \frac{1}{3} = \frac{1}{3} + \frac{1}{2} = \left(\frac{5}{6} \right)$$

Независ. $P(A \cap B) = P(A) \cdot P(B)$

Услов: $P(B|A) = \frac{P(A \cap B)}{P(A)}$

1.

Non.

Np

0,4

0,6

$$1 - 0,6^n \geq 0,9$$

←

$$0,6^n \leq 0,1$$

$$\left(\frac{6}{10}\right)^n \leq \frac{1}{10}$$

$$0,6^n \leq 0,1$$

$$\left(\frac{6}{10}\right)^n \leq \frac{1}{10}$$

$$n = 2$$

$$n = 3$$

$$n = 4$$

$$n = 5$$

$$0,6 \leq 0,1$$

$$0,36 \leq 0,1$$

$$0,216 \leq 0,1$$

$$0,1296 \leq 0,1$$

$$0,07776 \leq 0,1$$

2. 5 op nok
4 op nok

$$P(A) = \frac{N(A)}{N}$$

$$P(B) = \frac{N(B)}{N}$$

$$\frac{P(A)}{P(B)} = \frac{N(A)}{N(B)}$$

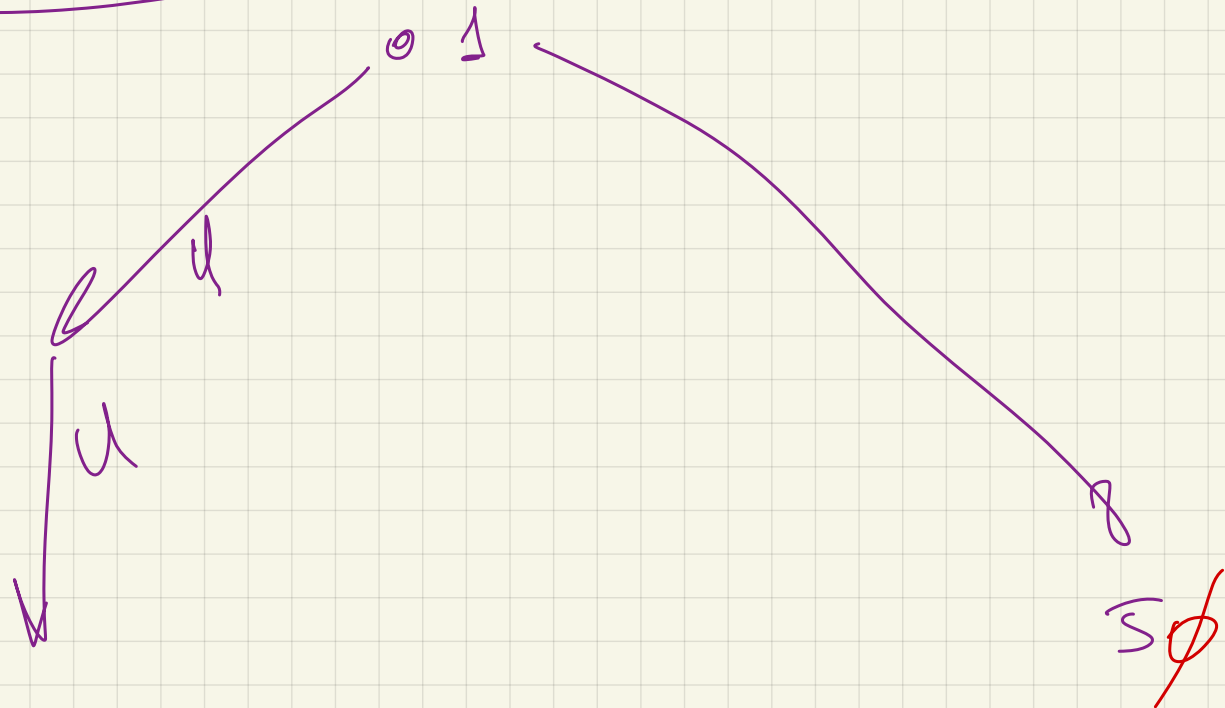
$$C_n^k = \frac{n!}{k!(n-k)!}$$

$$N(A) = C_{10}^5 = \frac{10!}{5!(10-5)!} = \frac{10!}{5! \cdot 5!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 252$$

$$N(B) = C_{10}^4$$

$$\frac{P(A)}{P(B)} = \frac{252}{210} = 1.2$$

> 4 $2 \sigma p$



$\frac{1}{12}$ $-\frac{1}{9}$ $\frac{5}{36}$ $\frac{6}{36}$

$$P(A) = 0,2$$

$$P(B) = 0,2$$

$$P(A \cap B) = 0,14$$

$$P(A \cup B) =$$

$$P(A) + P(B) - P(A \cap B)$$

$$= 0,2 + 0,2 - 0,14$$

$$0,26$$

0,74

$$P(1) = C_5^1 \cdot \underbrace{(1-0,2)}_{\text{огрм. сумм}} \cdot \underbrace{0,2 \cdot 0,2 \cdot 0,2 \cdot 0,2}_{\text{4 сервера по 0,2}}$$

$$P(3) = C_5^3 = 20 \cdot (1-0,2)(1-0,2)(1-0,2)0,2 \cdot 0,2$$

$$\frac{P(1)}{P(3)} = \frac{5}{20} \cdot \frac{\cancel{0,8} \cdot 0,2 \cdot \cancel{0,2} \cdot 0,2 \cdot 0,2}{\cancel{0,8} \cdot 0,8 \cdot 0,8 \cdot \cancel{0,2} \cdot \cancel{0,2}}$$

$$\frac{5}{20} \cdot \frac{2 \cdot 2}{8 \cdot 8} = \frac{20}{20 \cdot 8 \cdot 8} = \frac{1}{64}$$